Foundational Probability Concepts

Lecture 5 Notes on Conditional Probability & The Law of Total Probability

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1 Lecture 1: Introduction to Conditional Probability

1.1 The Core Idea: Updating Probabilities

In probability, we often want to answer the question: *How does our knowledge that one event, B, has occurred affect the probability of another event, A?* This is the central concept of conditional probability. It allows us to refine and update our probability estimates based on new, partial information.

Key Concept: Conditional Probability Notation

We write the probability of event A occurring, given that event B has occurred, as:

$$P(A|B)$$

This is read as "the probability of A, given B."

1.2 Visualizing Conditional Probability

A Venn diagram helps build intuition. The entire set of outcomes is the sample space, Ω. When we are **given** that event B has occurred, our world shrinks. We are no longer considering all of Ω; we can effectively "zoom in" on the region representing B.

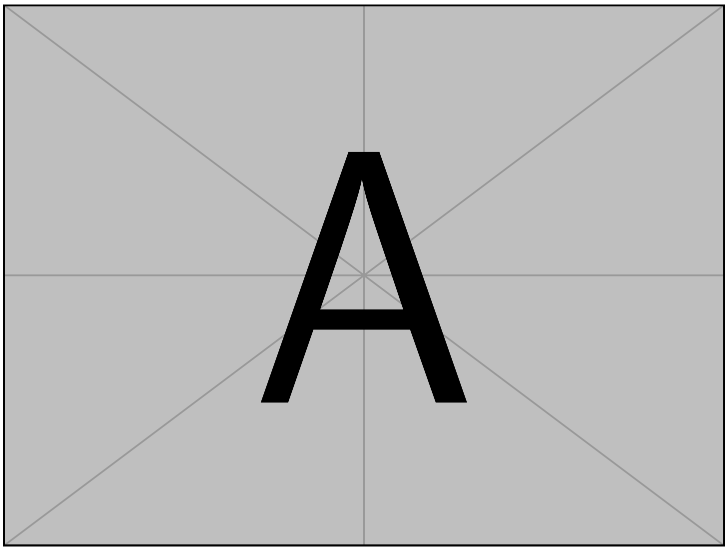

Figure 1: When we know B has happened, our sample space shrinks from Ω to just B. The probability of A happening now is the portion of A that is also in B (the intersection $A \cap B$) relative to the new total space, B.

The conditional probability $P(A|B)$ is the ratio of the area of the intersection ($A \cap B$) to the area of our new, smaller sample space (B).

Definition Formula: Conditional Probability

The formal definition of conditional probability is derived from this intuition:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Where:

- $P(A|B)$ is the probability of A occurring, given that B has occurred.
- $P(A \cap B)$ is the probability that **both** A and B occur (the joint probability).
- $P(B)$ is the probability of the given event B occurring. It must be that $P(B) > 0$.

1.3 Examples to Build Intuition

Example: Drawing Cards

We draw one card from a standard 52-card deck.

- Let Event A = The card is a Spade.
- Let Event B = The card is black.
- Let Event C = The card is red.

Baseline Probability: The unconditional probability of drawing a Spade is $P(A) = \frac{13}{52} = \frac{1}{4}$.
Case 1: Probability of A given B (the card is black) Knowing the card is black restricts our sample space to the 26 black cards. Of these, 13 are Spades.

$$P(A|B) = \frac{\text{Number of Spades}}{\text{Number of Black Cards}} = \frac{13}{26} = \frac{1}{2}$$

Knowing the card is black *doubled* the probability of it being a Spade.
Case 2: Probability of A given C (the card is red) Knowing the card is red restricts our sample space to the 26 red cards. Of these, 0 are Spades.

$$P(A|C) = \frac{\text{Number of Spades}}{\text{Number of Red Cards}} = \frac{0}{26} = 0$$

Knowing the card is red makes it *impossible* for it to be a Spade.

1.4 Real-World Application: Medical Screening

Application: Medical Test Interpretation

Conditional probability is the foundation of interpreting medical tests. Consider a new cancer test evaluated on 1,000 people.

Condition	**Test Positive**	**Test Negative**	**Total**
Has Cancer	450	50	500
No Cancer (Control)	100	400	500
Total	550	450	1000

Problem: What is the probability of testing positive, *given* that a person has cancer?
Solution:

- Let A = The test result is positive.
- Let B = The person has cancer.

We want to find $P(A|B)$. The condition "given that a person has cancer" means we restrict our sample space to the 500 people in that group. Within this group, 450 tested positive.

$$P(A|B) = \frac{\text{Number with cancer who test positive}}{\text{Total number with cancer}} = \frac{450}{500} = 0.90$$

This value, $P(\text{Positive Test}|\text{Disease})$, is the **sensitivity** of the test.

Looking Ahead

In the medical example, we calculated $P(A|B)$. However, a patient really wants to know the inverse: $P(B|A)$, the probability they have cancer given a positive test. This is a much harder question that requires **Bayes' Theorem**, which builds directly on the concepts discussed here.

Practice Problems

1. **Deck of Cards:** A single card is drawn from a standard 52-card deck. What is the probability that it is a Queen, given that it is a face card (Jack, Queen, or King)?

2. **Rolling Dice:** Two fair dice are rolled. Given that the sum of the dice is greater than 9, what is the probability that the first die was a 6?

3. **Surveys:** In a survey of 100 students, 60 like pizza and 45 like burgers. 20 students like both.

   a) What is the probability that a randomly selected student likes pizza, given that they like burgers?
   b) What is the probability that a randomly selected student likes burgers, given that they like pizza?

2 The Law of Total Probability

2.1 Review and First Corollary: The Multiplication Law

From the definition of conditional probability, we can derive a simple but powerful tool for finding the probability of an intersection of two events.

Definition Formula: The Multiplication Law

Starting with $P(A|B) = \frac{P(A \cap B)}{P(B)}$, we can rearrange it to find the joint probability:

$$P(A \cap B) = P(A|B) \times P(B)$$

In words: The probability of A and B both occurring is the probability of A occurring given B has occurred, multiplied by the probability that B occurs.

2.2 Second Corollary: The Law of Total Probability

This law provides a method for calculating the probability of an event by breaking the problem down into smaller, mutually exclusive pieces.

Key Concept: Partitioning the Sample Space

To use the law, we must first **partition** the sample space Ω into a set of events $B_1, B_2, \ldots, B_n$ that satisfy two conditions:

1. **Mutually Exclusive (Disjoint):** No two events can happen at the same time. The intersection of any two distinct events is the empty set ($B_i \cap B_j = \emptyset$ for $i \neq j$).
2. **Exhaustive:** The events cover all possible outcomes. Their union is the entire sample space ($B_1 \cup B_2 \cup \cdots \cup B_n = \Omega$).

For example, the four suits (Hearts, Diamonds, Clubs, Spades) form a partition of a deck of cards.

Definition Formula: The Law of Total Probability

If $B_1, B_2, \ldots, B_n$ form a partition of the sample space, then for any event A:

$$P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

Expanded, this is:

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_n)P(B_n)$$

The total probability of A is the weighted average of its conditional probabilities across all possible scenarios B_i.

Example: Drawing a Red Card

Problem: What is the probability of drawing a red card from a 52-card deck? (We know the answer is $\frac{1}{2}$, but let's prove it with the law).

Step 1. **Define Event and Partition:**

- Event A: The card is red.
- Partition B_i: The four suits, where B_1 = Hearts, B_2 = Diamonds, B_3 = Clubs, B_4 = Spades.

Step 2. **Probabilities of the Partition:** The probability of drawing a card from any suit is $\frac{1}{4}$. So, $P(B_1) = P(B_2) = P(B_3) = P(B_4) = \frac{1}{4}$.

Step 3. **Calculate Conditional Probabilities:**

- $P(A|B_1)$: Probability card is red, given it's a Heart. This is 1.
- $P(A|B_2)$: Probability card is red, given it's a Diamond. This is 1.
- $P(A|B_3)$: Probability card is red, given it's a Club. This is 0.
- $P(A|B_4)$: Probability card is red, given it's a Spade. This is 0.

Step 4. **Apply the Law:**

$$\begin{aligned}P(A) &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) + P(A|B_4)P(B_4)\\ &= (1 \times \frac{1}{4}) + (1 \times \frac{1}{4}) + (0 \times \frac{1}{4}) + (0 \times \frac{1}{4})\\ &= \frac{1}{4} + \frac{1}{4} + 0 + 0 = \frac{2}{4} = \frac{1}{2}\end{aligned}$$

2.3 A Common Special Case: B and not B

The simplest and most useful partition involves an event B and its complement, B^c ("not B"). The set $\{B, B^c\}$ is always a valid partition. In this case, the Law of Total Probability simplifies.

Key Concept: Special Case for Complements

For any two events A and B, the probability of A can be found by conditioning on whether B happens or not:

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

Summary & Key Takeaways

- **Multiplication Law:** Used to find the probability of a joint event.

$$P(A \cap B) = P(A|B) \times P(B)$$

- **Law of Total Probability:** Used to find the probability of an event A by summing its conditional probabilities over a partition of the sample space.

$$P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

- These two laws are essential building blocks for understanding and applying **Bayes' Theorem**.

Practice Problems

1. **Urn Problem:** Urn 1 contains 2 red and 3 blue balls. Urn 2 contains 4 red and 1 blue ball. You flip a fair coin. If it's heads, you draw a ball from Urn 1. If it's tails, you draw from Urn 2. What is the total probability of drawing a red ball?

2. **Manufacturing Defects:** A factory has two machines, X and Y. Machine X produces 60% of the daily output, and 3% of its products are defective. Machine Y produces the remaining 40%, and 5% of its products are defective. If you randomly select one product, what is the probability that it is defective?

3. **Weather and Commute:** The probability of rain on any given day is 0.3. If it rains, the probability you will be late for work is 0.4. If it does not rain, the probability you will be late is 0.1. What is the probability that you will be late for work tomorrow?

3 Solutions to Practice Problems

3.1 Lecture 1 Solutions

1. **Deck of Cards (Queen given Face Card):**
   - Let A = Card is a Queen.
   - Let B = Card is a face card (J, Q, K).
   - The total number of face cards in a deck is 3 suits $\times$ 4 cards/suit $= 12$. This is our new sample space.
   - The number of Queens in the deck is 4. All of these are also face cards.
   - Therefore, $P(A|B) = \frac{\text{Number of Queens}}{\text{Number of Face Cards}} = \frac{4}{12} = \frac{1}{3}$.
2. **Rolling Dice (First die is 6 given sum > 9):**
   - Let A = First die is a 6.
   - Let B = Sum of dice is greater than 9 (i.e., 10, 11, or 12).
   - First, list all outcomes where the sum is greater than 9. This is our new sample space:
     - Sum of 10: (4,6), (5,5), (6,4)
     - Sum of 11: (5,6), (6,5)
     - Sum of 12: (6,6)

     There are 6 possible outcomes in our new sample space.
   - Now, count how many of these outcomes have a 6 on the first die: (6,4), (6,5), (6,6). There are 3 such outcomes.
   - Therefore, $P(A|B) = \frac{\text{Outcomes where first die is 6}}{\text{Total outcomes where sum > 9}} = \frac{3}{6} = \frac{1}{2}$.
3. **Surveys (Pizza and Burgers):**
   - Let Pz = Student likes pizza. $P(Pz) = 60/100 = 0.6$.
   - Let B = Student likes burgers. $P(B) = 45/100 = 0.45$.
   - Let $Pz \cap B$ = Student likes both. $P(Pz \cap B) = 20/100 = 0.2$.

   a) **Likes pizza given likes burgers:** We want $P(Pz|B)$.

   $$P(Pz|B) = \frac{P(Pz \cap B)}{P(B)} = \frac{0.2}{0.45} = \frac{20}{45} = \frac{4}{9} \approx 0.444$$

   b) **Likes burgers given likes pizza:** We want $P(B|Pz)$.

   $$P(B|Pz) = \frac{P(Pz \cap B)}{P(Pz)} = \frac{0.2}{0.6} = \frac{20}{60} = \frac{1}{3} \approx 0.333$$

3.2 Lecture 2 Solutions

1. **Urn Problem:**

   - Let A = Draw a red ball.
   - Let B_1 = Choose Urn 1 (Heads). $P(B_1) = 0.5$.
   - Let B_2 = Choose Urn 2 (Tails). $P(B_2) = 0.5$.
   - The set $\{B_1, B_2\}$ forms a partition of the sample space.
   - Conditional Probabilities:
     - $P(A|B_1)$: Probability of red, given Urn 1. There are 2 red out of 5 balls. $P(A|B_1) = \frac{2}{5}$.
     - $P(A|B_2)$: Probability of red, given Urn 2. There are 4 red out of 5 balls. $P(A|B_2) = \frac{4}{5}$.
   - Apply the Law of Total Probability:

   $$\begin{aligned} P(A) &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) \\ &= \left(\frac{2}{5} \times 0.5\right) + \left(\frac{4}{5} \times 0.5\right) \\ &= \frac{1}{5} + \frac{2}{5} = \frac{3}{5} = 0.6 \end{aligned}$$

   The total probability of drawing a red ball is 60%.

2. **Manufacturing Defects:**

   - Let A = Product is defective.
   - Let B_X = Product is from Machine X. $P(B_X) = 0.6$.
   - Let B_Y = Product is from Machine Y. $P(B_Y) = 0.4$.
   - The set $\{B_X, B_Y\}$ is a partition.
   - Conditional Probabilities (given in the problem):
     - $P(A|B_X)$ = 0.03 (3% defect rate for Machine X).
     - $P(A|B_Y)$ = 0.05 (5% defect rate for Machine Y).
   - Apply the Law of Total Probability:

   $$\begin{aligned} P(A) &= P(A|B_X)P(B_X) + P(A|B_Y)P(B_Y) \\ &= (0.03 \times 0.6) + (0.05 \times 0.4) \\ &= 0.018 + 0.020 = 0.038 \end{aligned}$$

   The overall probability of selecting a defective product is 3.8%.

3. **Weather and Commute:**

   - This is a direct application of the special case with complements.
   - Let A = You are late for work.
   - Let B = It rains. $P(B) = 0.3$.

- The complement is B^c = It does not rain. $P(B^c) = 1 - P(B) = 1 - 0.3 = 0.7$.
- Conditional Probabilities:
  - $P(A|B) = 0.4$ (Probability of being late, given rain).
  - $P(A|B^c) = 0.1$ (Probability of being late, given no rain).
- Apply the Law of Total Probability:

$$\begin{aligned} P(A) &= P(A|B)P(B) + P(A|B^c)P(B^c) \\ &= (0.4 \times 0.3) + (0.1 \times 0.7) \\ &= 0.12 + 0.07 = 0.19 \end{aligned}$$

  The probability that you will be late for work tomorrow is 19%.